

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed **biodegradable material formulations** for flexible polymer substrates integrated with **direct laser writing (DLW)** fabrication processes, achieving porous electrode structures suitable for wearable electrochemical sensors in the e-FLOWER project.
- Characterized **laser-induced graphene (LIG)** materials and 3D conductive electrode architectures using **scanning electron microscopy (SEM)**, **Raman spectroscopy**, and **electrochemical impedance spectroscopy (EIS)**, validating performance metrics for energy storage and sensing applications.
- Developed and optimized **direct laser writing** fabrication protocols for manufacturing e-textile components (sensors and supercapacitors) on natural polymer substrates, documenting process parameters and yield across **25+ experimental iterations**.
- Conducted **electrochemical characterization** and **cyclic voltammetry (CV)** testing on fabricated LIG electrode materials to validate charge-storage capacity, cycle stability, and biocompatibility for land, water, and space deployment environments.
- Managed experimental design, data collection, and **statistical analysis** using **R / RStudio** and **Python (pandas, NumPy, scikit-learn)** for material performance evaluation and fabrication process optimization across wearable sensor and energy-storage components.
- Synthesized technical findings into **peer-reviewed research documentation**, **conference presentations**, and **IP disclosures** on sustainable, biodegradable e-textile technologies; collaborated with cross-functional research teams on materials science and device integration methodologies.

SKILLS

Materials Science & Fabrication – laser-induced graphene (LIG), direct laser writing (DLW), biodegradable polymers, flexible substrates, porous electrode design, natural materials sourcing, 3D material characterization.

Analytical & Characterization Techniques – scanning electron microscopy (SEM), Raman spectroscopy, electrochemical impedance spectroscopy (EIS), cyclic voltammetry (CV), X-ray diffraction (XRD), thermal gravimetric analysis (TGA), tensile testing.

E-Textile & Wearable Sensor Development – electrochemical sensors, energy-storage devices, supercapacitors, wearable component fabrication, biocompatibility assessment, stretchable electronics, sustainable wearable materials.

Data Analysis & Experimental Design – statistical analysis, R / RStudio, Python (pandas, NumPy, scikit-learn), experimental protocol design, process optimization, performance metrics evaluation, hypothesis testing.

Research & Technical Documentation – peer-reviewed publications, conference presentations, IP disclosures, lab notebook documentation, methodology standardization, research reporting, cross-functional collaboration.

Project Management & Sustainability – EU project coordination (e-FLOWER), circular economy principles, biodegradable material lifecycle assessment, environmental impact analysis, regulatory compliance for wearable medical devices.

EDUCATION AND TRAINING

PROJECTS

Biodegradable LIG Electrode Fabrication for Circular E-Textiles (e-FLOWER Project)

- Designed and synthesized **biodegradable material formulations** from natural polymer sources and developed **direct laser writing (DLW)** protocols to fabricate **laser-induced graphene (LIG)** electrodes on flexible substrates suitable for wearable sensors and energy-storage components.
- Performed comprehensive **electrochemical characterization** using **cyclic voltammetry**, **electrochemical impedance spectroscopy (EIS)**, and **charge-retention testing** to validate electrode performance, achieving **80%+ cycle stability** across **500+ charge-discharge cycles**.
- Applied **statistical analysis** and **Python-based data pipelines** (pandas, NumPy, scikit-learn) to process sensor-output data, optimize fabrication parameters, and model material degradation kinetics across land, water, and space deployment scenarios.

Multi-Modal Sensor Integration & Wearable Device Testing Platform (HCL Technologies / e-Textile Research)

- Engineered end-to-end testing framework for **wearable electrochemical sensors** and **energy-storage components**, integrating **SEM imaging**, **Raman spectroscopy**, and **thermal analysis (TGA)** to assess biocompatibility, mechanical durability, and environmental stability.
- Developed **automated data-acquisition workflows** using **Python** to process and correlate material characterization results, reducing manual analysis time by **40%** and enabling rapid iteration on material formulations.
- Documented **fabrication protocols**, **material safety data sheets (MSDS)**, and **experimental methodologies** for reproducibility; presented findings in **peer-reviewed publications** and **EU project consortium meetings** on sustainable wearable technologies.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)